



11618 - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Cycle: 5, Proposal Category: GO

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Observation Folder				
	1	MRS-A	MIRI Medium Resolution Spectroscopy	(1) DEEP23-z2LRD1-MRS
	4	MRS-C	MIRI Medium Resolution Spectroscopy	(1) DEEP23-z2LRD1-MRS
	5	PRISM	NIRSpec IFU Spectroscopy	(3) DEEP23-z2LRD1-PRISM
	2	Background A	MIRI Medium Resolution Spectroscopy	(2) DEEP23-z2LRD1-SKY
	3	Background C	MIRI Medium Resolution Spectroscopy	(2) DEEP23-z2LRD1-SKY

ABSTRACT

JWST's discovery of a new class of broad-line objects at $2 < z < 9$, little red dots (LRDs), has sparked great interest with their nature being hotly debated. Proposed interpretation range from dense stellar populations, to dust-obscured AGN, to gas-enriched accreting black holes. Whenever a new wavelength regime is opened up for LRDs, they always surprise us and motivate novel interpretations. Therefore, gaining a comprehensive SED coverage is crucial to understand their nature. With many JWST observation in the rest UV-to-NIR and no detection in the X-ray, FIR, and radio, the MIR (rest 4-8.5 microns) is now the key uncharted regime where dust and host galaxy could be detected. This wavelength range is unlocked at $z \sim 2$, and only JWST is able to reach it. Leveraging the wide wavelength coverage of NIRSpec and MIRI, we propose NIRSpec/PRISM+MIRI/MRS spectroscopy for the brightest little red dot, DEEP23-z2LRD1, at cosmic noon to gain a full UV-to-MIR (rest 0.3-8.5 micron) spectroscopic mapping of an LRD. By disentangling the continuum emission with any high-equivalent-width features, we will characterize the origin of the previously uncharted MIR SED. A full UV-to-optical spectroscopy will put key constraints on the dust, the central engine, and/or the host galaxy properties of LRDs, all of which are poorly understood. Such wide wavelength coverage will allow us to directly measure bolometric luminosity estimates, which is key to understand the accretion power of the central engine of LRDs. This program, will deliver, for the first time, composite models from UV to MIR for the LRDs to motivate future theory development and observations.

OBSERVING DESCRIPTION

This program aims to obtain NIRSpec/PRISM and MIRI/MRS observation of the brightest little red dot at cosmic noon, DEEP23-z2LRD1, to gain the first rest-frame UV-to-MIR (rest 0.3-8.5 microns) spectroscopic coverage of a luminous LRD.

For NIRSpec/PRISM observation, we plan to use a configuration of 5 groups/integration, 1 integration/exposure, the NRS readout pattern, and a 8-point dither pattern. This makes a total of 1803s of exposure time to obtain high signal-to-noise rest-frame UV-to-NIR coverage.

For the MRS observation, we will only use the SHORT(A) and LONG(C) sub-bands to cover key diagnostic spectral features such as the Brackett lines and the PAHs, while ensuring required continuum signal-to-noise ratio. For each sub-bands, we will use a configuration of 100 groups/integration, 13 integrations/exposure, a FASTR1 readout pattern, and a 4-point dither pattern. This makes a total of 29 ks of on-source observation for both-bands. The MRS observation will require acquisition from a nearby Gaia source.

The MRS observation also requires separate background observations executed on a non-interruptible sequence to mitigate persistence. We will use a configuration of 100 groups/integration, 13 integrations/exposure, a FASTR1 readout pattern, and a 2-point dither pattern. This will total of 14.5 ks of exposure time for both sub-bands. While taking background spectroscopy, we will plan simultaneous imaging of F2100W (during A sub-band background exposure) and F2550W (during C sub-band background exposure) on our science target, with a configuration of 13 groups/integration, 81 integrations per exposure, FASTR1 readout mode, and 2-point dither pattern. Each band totals 6299s of exposure time. The shorter imaging exposure time than that of the background spectroscopy is to avoid background saturation.

Proposal 11618 - Targets - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	DEEP23-z2LRD1-MRS	RA: 23 27 35.8800 (351.8995000d) Dec: -00 13 41.52 (-.22820d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Active galactic nuclei]</i> <i>Extended=NO</i>		
	(2)	DEEP23-z2LRD1-SKY	RA: 23 27 35.8800 (351.8995000d) Dec: -00 13 41.52 (-.22820d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Active galactic nuclei]</i> <i>Extended=NO</i>		
	(3)	DEEP23-z2LRD1-PRISM	RA: 23 27 35.8800 (351.8995000d) Dec: -00 13 41.52 (-.22820d) Equinox: J2000 <i>Comments:</i> <i>Category=Galaxy</i> <i>Description=[Active galactic nuclei]</i> <i>Extended=NO</i>		
	(4)	Gaia_Star	RA: 23 27 35.8935 (351.8995562d) Dec: -00 13 52.02 (-.23112d) Equinox: J2000 <i>Comments:</i> <i>Category=Star</i> <i>Description=[G stars]</i> <i>Extended=NO</i>	Epoch of Position: 2000	

Proposal 11618 - Observation 1 - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Observation	Proposal 11618, Observation 1: MRS-A													Fri Mar 13 21:08:52 GMT 2026
	Diagnostic Status: Warning													
	Observing Template: MIRI Medium Resolution Spectroscopy													
	Background Observations:[Background A (Obs 2)]													
Diagnostics	(MRS-A (Obs 1)) Warning (Form): Filter mismatch between science and background observations may result in incorrect background subtraction.													
	(MRS-A (Obs 1)) Warning (Form): Observers are responsible for checking that target acquisition is feasible.													
	(Visit 1:1) Warning (Form): Data Excess over lower threshold													
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.													
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous			
	(1)	DEEP23-z2LRD1-MRS	RA: 23 27 35.8800 (351.8995000d) Dec: -00 13 41.52 (-.22820d) Equinox: J2000											
	Comments: Category=Galaxy Description=[Active galactic nuclei] Extended=NO													
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Exposure Time	Optional ETC ID				
	1	4 Gaia_Star	F560W	FAST	10	1		1	27.75					
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction					
	All MRS		YES			FULL			Allow Auto Reorder					
Dithers	#	Dither Type				Optimized For				Direction				
	1	4-Point				POINT SOURCE				NEGATIVE				
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1		IMAGER	F770W	SLOWR1	20	7	1	Dither 1	4	28	13951.713		
	1	SHORT(A)	MRSLONG		FASTR1	100	13	1	Dither 1	4	52	14563.41		
	1	SHORT(A)	MRSSHORT		FASTR1	100	13	1	Dither 1	4	52	14563.41		

Proposal 11618 - Observation 1 - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 11618 - Observation 4 - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Observation	Proposal 11618, Observation 4: MRS-C												Fri Mar 13 21:08:52 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observations:[Background C (Obs 3)]												
Diagnostics	(MRS-C (Obs 4)) Warning (Form): Filter mismatch between science and background observations may result in incorrect background subtraction.												
	(MRS-C (Obs 4)) Warning (Form): Observers are responsible for checking that target acquisition is feasible.												
	(Visit 4:1) Warning (Form): Data Excess over lower threshold												
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections				Miscellaneous		
	(1)	DEEP23-z2LRD1-MRS	RA: 23 27 35.8800 (351.8995000d) Dec: -00 13 41.52 (-.22820d) Equinox: J2000										
	Comments: Category=Galaxy Description=[Active galactic nuclei] Extended=NO												
Acquisition	#	Target	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID				
	1	4 Gaia_Star	F560W	FAST	10	1	1	27.75					
Template	Primary Channel		Simultaneous Imaging			Imager Subarray			Grating Wheel Direction				
	All MRS		YES			FULL			Allow Auto Reorder				
Dithers	#	Dither Type				Optimized For				Direction			
	1	4-Point				POINT SOURCE				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F1000W	SLOWR1	20	7	1	Dither 1	4	28	13951.713	
	1	LONG(C)	MRSLONG		FASTR1	100	13	1	Dither 1	4	52	14563.41	
	1	LONG(C)	MRSSHORT		FASTR1	100	13	1	Dither 1	4	52	14563.41	

Proposal 11618 - Observation 4 - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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Proposal 11618 - Observation 5 - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Observation	Proposal 11618, Observation 5: PRISM											Fri Mar 13 21:08:52 GMT 2026
	Diagnostic Status: Warning											
	Observing Template: NIRSpec IFU Spectroscopy											
Diagnostics	(PRISM (Obs 5)) Warning (Form): Observers are responsible for checking that target acquisition is feasible.											
	(Visit 5:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(3)	DEEP23-z2LRD1-PRISM	RA: 23 27 35.8800 (351.8995000d) Dec: -00 13 41.52 (-.22820d) Equinox: J2000									
	Comments: Category=Galaxy Description=[Active galactic nuclei] Extended=NO											
Acquisition	#	Target	TA Method	Subarray	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	3 DEEP23-z2LRD1-PRISM	WATA	FULL	CLEAR	NRSRAPID	3	1	1	42.947		
Template	HFF Readout Mode											
	false											
Dithers	#	Dither Type			Size		Starting Point		Number of Points		Points	
	1	CYCLING			SMALL		1		8			
Spectral Elements	#	Grating/Filter	Readout Pattern	Groups/Int	Integrations/Exp	Leakcal	Dither	Autocal	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1	PRISM/CLEAR	NRS	5	1	false	true	NONE	8	8	1803.777	

Proposal 11618 - Observation 2 - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Observation	Proposal 11618, Observation 2: Background A												Fri Mar 13 21:08:52 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [MRS-A (Obs 1)]												
Diagnostics	(Visit 2:1) Warning (Form): Data Excess over lower threshold												
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(2)	DEEP23-z2LRD1-SKY	RA: 23 27 35.8800 (351.8995000d) Dec: -00 13 41.52 (-.22820d) Equinox: J2000										
	Comments: Category=Galaxy Description=[Active galactic nuclei] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel			Simultaneous Imaging			Imager Subarray		Grating Wheel Direction			
	F1500W	Imager			YES			FULL		Allow Auto Reorder			
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F2100W	FASTR1	13	81	1	Dither 1	2	162	6288.241	
	1	SHORT(A)	MRSLONG		FASTR1	100	13	1	Dither 1	2	26	7281.705	
	1	SHORT(A)	MRSSHORT		FASTR1	100	13	1	Dither 1	2	26	7281.705	

Proposal 11618 - Observation 2 - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Special Requirements	Sequence Observations 1, 2, Non-interruptible
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Proposal 11618 - Observation 3 - UV-to-MIR Spectroscopy of the Brightest Little Red Dot at Cosmic Noon

Observation	Proposal 11618, Observation 3: Background C												Fri Mar 13 21:08:52 GMT 2026
	Diagnostic Status: Warning												
	Observing Template: MIRI Medium Resolution Spectroscopy												
	Background Observation For: [MRS-C (Obs 4)]												
Diagnostics	(Visit 3:1) Warning (Form): Data Excess over lower threshold												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(2)	DEEP23-z2LRD1-SKY	RA: 23 27 35.8800 (351.8995000d) Dec: -00 13 41.52 (-.22820d) Equinox: J2000										
	Comments: Category=Galaxy Description=[Active galactic nuclei] Extended=NO												
Acquisition	#	Target											
	1	NONE											
Template	AcqFilter	Primary Channel				Simultaneous Imaging			Imager Subarray		Grating Wheel Direction		
	F1500W	Imager				YES			FULL		Allow Auto Reorder		
Dithers	#	Dither Type				Optimized For				Direction			
	1	2-Point				BACKGROUND				NEGATIVE			
Spectral Elements	#	Wavelength Range	Detector	Filter	Readout Pattern	Groups/Int	Integrations/Exp	Exposures/Dith	Dither	Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID
	1		IMAGER	F2550W	FASTR1	13	81	1	Dither 1	2	162	6288.241	
	1	LONG(C)	MRSLONG		FASTR1	100	13	1	Dither 1	2	26	7281.705	
	1	LONG(C)	MRSSHORT		FASTR1	100	13	1	Dither 1	2	26	7281.705	

Special Requirements	Sequence Observations 3, 4, Non-interruptible
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